PredMainAI — Phase 3 MVP Report

Group 3 - MAE 301
Aidan Avona, Seth Christman, Advik Kaura, AL AYHAM AL HADHRAMI, Ethan Gilletti

1. Executive Summary

Industrial machines often fail unexpectedly, leading to costly downtime and maintenance. Traditional maintenance approaches either react too late or rely on rigid rule-based thresholds that fail to capture complex failure patterns.

PredMain AI is a predictive maintenance system that analyzes machine sensor data to detect anomalies and predict failures before they occur. The MVP allows users to upload a dataset of sensor readings and receive:

- Failure predictions across multiple models
- Machine health scores
- Risk rankings of machines
- Visual insights into failure drivers

The system combines anomaly detection (Isolation Forest, Autoencoder) with sequence modeling (LSTM) to provide both detection and interpretability.

2. User & Use Case

Target User

- Maintenance engineer in a manufacturing plant

Use Case Scenario

A maintenance engineer uploads a CSV file containing machine sensor data (temperature, torque, wear, etc.). The system processes the data and outputs:

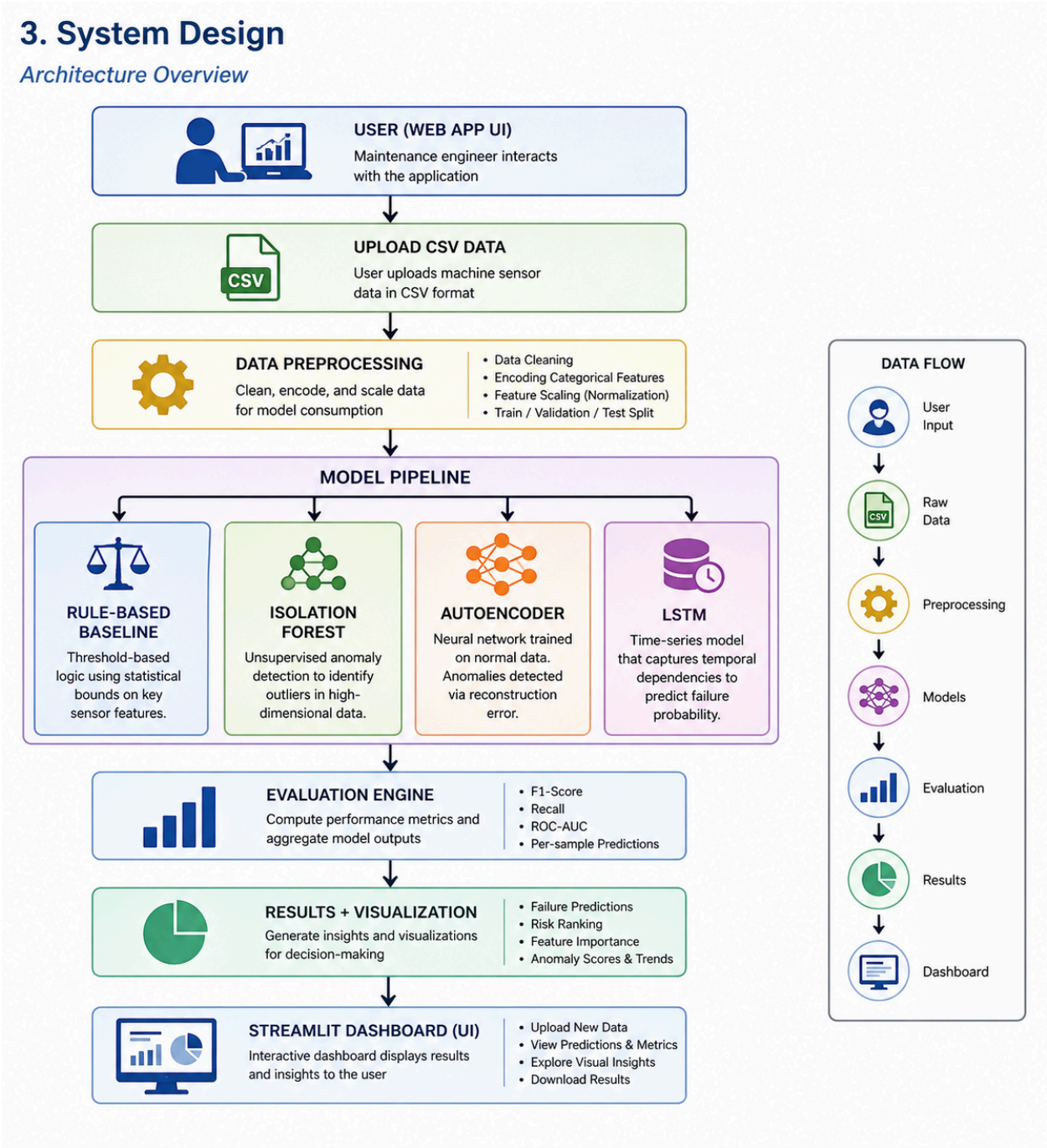
- Which machines are likely to fail
- A ranked list of high-risk machines
- Explanations of why failures are predicted
- Visual insights into system behavior

This allows the engineer to:

- prioritize maintenance
- reduce downtime

- act proactively instead of reactively

3. System Design



Key Design Points

- Streamlit serves as the frontend interface
- Models are executed in a unified pipeline

- Results are aggregated into a single DataFrame
- Outputs include both predictions and explanations

4. Data

Dataset Used

- AI4I 2020 Predictive Maintenance Dataset (UCI)

Size

- ~10,000 samples
- ~9 features

Features

- Air temperature
- Process temperature
- Rotational speed
- Torque
- Tool wear
- Machine type

Preprocessing Steps

- Renamed columns to standardized format
- Encoded categorical variables (Type → numeric)
- Normalized numerical features (StandardScaler)
- Created binary target (failure_label)

Data Split

- Train: 70%
- Validation: 15%
- Test: 15%
- Stratified by failure class

5. Models

The system uses a combination of baseline and AI models:

Rule-Based Baseline

- Threshold-based detection
- Flags failure when:

    - torque or speed exceeds statistical bounds

Isolation Forest

- Unsupervised anomaly detection
- Trained only on normal samples
- Detects deviations from typical machine behavior

Autoencoder (Neural Network)

- Learns compressed representation of normal data
- Flags anomalies based on reconstruction error

LSTM (Time-Series Model)

- Uses sliding window of sensor data
- Captures temporal patterns in machine behavior
- Predicts failure probability over time

Model Strategy

- Hybrid approach:
    - statistical baseline
    - unsupervised anomaly detection
    - deep learning model

This improves robustness across different failure patterns.

6. Evaluation

Quantitative Results (Example)

Model	F1 Score	Recall	ROC-AUC	Verdict
Isolation Forest	0.485	0.941	0.934	Decent
LSTM	0.381	0.235	0.923	Weak
Autoencoder	0.301	1.000	0.519	Weak
Rule-Based	0.296	0.235	0.611	Weak

Key Insights

- Isolation Forest performs best overall
- Autoencoder detects all failures but produces many false alarms

- LSTM underperforms due to limited temporal structure in data
- High tool wear strongly correlates with failures
- Elevated torque + wear is a key failure pattern
- Some false positives show early warning signals rather than errors

7. Limitations & Risks

Technical Limitations

- LSTM performance limited by lack of strong time-series structure
- High false positive rate in anomaly models
- Duplicate column issues required post-processing fixes

Data Limitations

- Class imbalance (~97% normal)
- Synthetic-like dataset may not reflect real industrial variability

Risks

- False alarms → unnecessary maintenance
- Missed failures → critical system breakdown
- Model generalization may be limited to similar datasets

8. Next Steps

Technical Improvements

- Improve LSTM with better sequence engineering
- Tune anomaly thresholds to reduce false positives
- Add feature importance (e.g., SHAP values)

Product Improvements

- Deploy fully on Streamlit Cloud
- Add real-time monitoring dashboard
- Support live sensor streaming

Advanced Expansion

- Integrate with IoT systems
- Add predictive scheduling (when failure will occur)
- Build API for industrial integration

Summary

PredMain AI successfully demonstrates a working predictive maintenance MVP that:

- detects failures
- ranks machine risk
- provides interpretable insights

While performance can be improved, the system delivers a strong foundation for real-world deployment and further development.